

- ✓ Information Technology Infrastructure Library
- ✓ ITIL is a framework of ITSM (Information Technology Service Management).
- ✓ ITSM consists of various processes, policies and procedures for aiming better designs, delivery, management and improving the way IT services are utilized within an organization.
- ✓ ITIL provides best practice and techniques for:
 - Selecting
 - Planning
 - Delivery
 - Maintaining
- ✓ Versions of ITIL:
 - ITIL v1 → 1989
 - ITIL v2 → 2001
 - ITIL v3 → 2007
 - ITIL v4 → 2019

Key concepts of ITIL:

- **Service lifecycle**
 - Service strategy – planning
 - Service design – here the planning transforms into processes, technology and architecture.
 - Service transition – ensures that new changes are built, tested and deployed into live environment successfully.
 - Service Operation – manage day-to-day services.
 - Continual service Improvement (CSI) – provides a mechanism for improving all aspects of IT services and processes used here.
- **Processes**
 - Incident management – restoring normal service operation as quickly as possible after an incident.
 - Change management – managing the lifecycle of all changes to minimize disruption.
 - Problem management – identifying and managing the root causes of incident.
 - Service Level Management – Ensuring that agreed-up service levels are met.
 - Capacity management – ensuring that IT infrastructure meets current and future capacity needs.
- **Functions**
 - Functions provide structure and stability to the lifecycle:
 - Service Desk – a single point of contact for users to communicate with the IT dept.
 - Technical management – manages the infra and provide the technical expertise.
 - Application management – manages applications throughout their lifecycle.
 - IT Operations management – manage day-to-day operations of the IT infra.
- **Roles and Responsibilities:**
 - ITIL defines various roles to ensure accountability and clarity.
 - Service Owner – responsible for overall design, performance, integration and improvement.
 - Process Owner – Ensure that a process is fit for its purpose and is responsible for process design, performance and improvements.
 - Service Manager – manages the development and delivery of the IT services.
 - IT Operations Manager – ensures efficient and effective delivery of IT services.

Basic concepts of ITILv4:

- Service Value System (SVS)
 - This describes “how all components and activities of an organization work together”.
- Service Value Chain (SVC)
 - It’s the central element of SVS to deliver, plan, improve, design and transition.
- Guiding Principles
 - These are universal recommendations to guide the organization.
- Four Dimensions of Service Management
 - Organization and people
 - Take care of roles & responsibilities
 - Information and technology
 - Data, tools, applications and infra.
 - Partners and suppliers
 - External parties that help deliver services.
 - Value streams and processes
 - How activities are organized to create values.
- Service Value System (SVS) Components
 - Guiding principles
 - Governance
 - SVC
 - Practices
 - Continual improvements.
- ITIL Practices
 - 34 practices
- Continual Improvement
 - Ensures services, processes and performance are always getting better.

Incident management

- ✓ Incident management is a process of restoring normal service operations as quick as possible.
- ✓ It focuses on quick resolution.
- ✓ Objectives of Incident management:
 - Detect and record incidents quickly.
 - Categorize and prioritize incidents for efficient handling.
 - Resolve incident in the shortest time possible.
 - Escalate incidents when necessary.
 - Keep users informed about the status & progress.
 - Reduce service downtime.
- ✓ Incident management applies to:
 - Hardware crash (server crash)
 - Software failure (application not working)
 - Network outage (internet is down)
 - User issues (password reset)
 - Performance degradation (app running slow)
- ✓ Basic Incident management workflow:
 - **Incident identification** – detect the incident via user, tool or alerts.
 - **Incident logging** – record (in detailed) the incident using ITSM tool.
 - **Categorization** – assign a category for reporting and analysis.
 - **Prioritization** – based on impact & urgency. (P1, P2, P3, P4)
 - **Initial diagnosis** – service desk attempts first-line resolution.
 - **Escalation** – passed to specialized team.
 - **Investigation and diagnosis** – a team will try to get the workaround.
 - **Resolution & recovery** – implementing the fix and confirmation of service restoration.
 - **Incident closure** – confirm with the user and update the ticket and close incident.
- ✓ Roles in incident management
 - Service desk
 - Incident Manager
 - Technical teams
 - Users

- ✓ Problem Management is a process focused on identifying, analyzing and resolving the root causes of recurring incidents or problems within the infrastructure.
- ✓ Key differences:
 - Incident management – fixes the symptoms quickly to restore the services.
 - Problem Management – finds and fix the root cause to prevent recurrences.
- ✓ Objectives
 - Identify and log problems.
 - Investigate the root cause of one or more incidents.
 - Develop workaround to minimize impact until a permanent fix is applied.
 - Create and maintain known error records for future resolution.
 - Prevent problems from happening again.
- ✓ Basic Problem Management workflow:
 - **Problem identification** – from related incidents, monitoring, notifications.
 - **Problem logging** – record details via ITSM tool.
 - **Problem categorization and prioritization** – this helps in allocating resources
 - **Problem investigation and diagnosis (RCA)**
 - 5 whys
 - Fishbone diagram
 - Fault tree analysis
 - **Workaround creation** – temporary solution
 - **Know Error Record creation** – a known error is a problem with a documented root cause and workaround.
 - **Problem resolution** – implanting permanent fix
 - **Problem closure** – ensure the fix is verified, update the KEDB (Known Error Database) and close the record.

Change management

- ✓ Change management is now called “Change Enablement”.
- ✓ The purpose of change enablement is to maximize the number of successful changes by ensuring that risks have been properly assessed.
- ✓ It ensures that changes to services, system or configurations are:
 - Planned
 - Evaluated
 - Approved
 - Implemented
 - Reviewed
- ✓ Objective of Change management:
 - Ensure that the changes are properly assessed for risk, impact and benefits.
 - Authorize only those changes that provide business values.
 - Minimize the negative effects on services.
 - Support fast and safe delivery of changes.
 - Maintain a schedule for change implementation.
- ✓ Scope for the change management:
 - Infrastructure changes (servers, networks, storage)
 - Application changes (new release, patches)
 - Process changes (new workflow, policy updates)
 - Configuration changes (set up new firewall rules, database)
- ✓ Types of changes:
 - Standard change
 - Low-risk, pre-approved changes.
 - Normal change
 - Goes through full assessment, authorization and scheduling.
 - It can be low, medium-risk or high.
 - Emergency change
 - Must be implemented urgently due to critical incident.
- ✓ Change management workflow:
 - **Request for change (RFC)** – RFC will be submitted by an authorized person using ITSM tool.
 - **Logging and categorizing** – identify the change type (standard, normal or emergency)
 - **Assessment and planning** – risk analysis, impact analysis & resource planning.
 - **Authorization** – approval by change authority
 - **Scheduling** – schedule when the changes are to be done.
 - **Implementation** – follow approved plans
 - **Review and closure** – confirm changes achieved and document lessons learned.